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Problem 29.

$S = C[0,1]$ and $d_p(f,g) \equiv (\int_0^1 |f(t)-g(t)|^p dt)^{\frac{1}{p}}$ for $1 \le p \le +\infty$ and $d_\infty(f,g) = \sup\{|f(t)-g(t)| : t \in [0,1]\}$

(a) $f(x) \equiv 1$, $f_n(t) = \begin{cases} 1 & 0 \le t \le 1-\frac{1}{n} \\ n(1-t) & 1-\frac{1}{n} \le t \le 1 \end{cases}$

$\Rightarrow \quad f(t) - f_n(t) = \begin{cases} 0 & 0 \le t \le 1-\frac{1}{n} \\ 1-n(1-t) & 1-\frac{1}{n} \le t \le 1 \end{cases} \qquad n = 1,2,3,\dots$

$1-n(1-t) \ge 0 \Rightarrow 1 \ge n(1-t) \Rightarrow t \ge 1-\frac{1}{n}$

so when $t \ge 1-\frac{1}{n}$ $\quad f(t)-f_n(t) = 1-n(1-t) \ge 0$

when $p < +\infty$

$$\begin{aligned}
\Rightarrow d_p(f,f_n) &= (\int_0^1 |f(t)-f_n(t)|^p dt)^{\frac{1}{p}} \\
&= (\int_0^{1-\frac{1}{n}} 0^p dt + \int_{1-\frac{1}{n}}^1 (1-n(1-t))^p dt)^{\frac{1}{p}} \\
&= (0 + \frac{[1-n(1-t)]^{p+1}}{p+1} \times \frac{1}{n} \Big|_{t=1-\frac{1}{n}}^{t=1})^{\frac{1}{p}} \\
&= (\frac{1}{p+1} \times \frac{1}{n} - \frac{(1-n\times\frac{1}{n})^{p+1}}{p+1} \times \frac{1}{n})^{\frac{1}{p}} \\
&= (\frac{1}{p+1} \times \frac{1}{n})^{\frac{1}{p}}
\end{aligned}$$

$\Rightarrow \lim_{n\to+\infty} d_p(f,f_n) = (\frac{1}{\infty})^{\frac{1}{p}} = 0$

when $p = +\infty$.

$$\begin{aligned}
d_\infty(f,f_n) &= \sup\{|f(t)-f_n(t)| : t \in [0,1]\} \\
&= \sup\{1-n(1-t) : t \in [1-\frac{1}{n}, 1]\} \\
&= 1 \quad \text{for all } n \ge 1
\end{aligned}$$

$\Rightarrow d_\infty(f,f_n) \longrightarrow 1 \ne 0$ as $n \to +\infty$.

(b) $g_n(t) - f(t) = \begin{cases} 0 & 0 \le t \le 1-\frac{1}{n} \\ f(1-\frac{1}{n}) + (f(1)-f(1-\frac{1}{n})) \times n \times (t+\frac{1}{n}-1) - f(t) & 1-\frac{1}{n} \le t \le 1 \end{cases} \qquad (*)$

$$\begin{aligned}
(*) &= f(1-\tfrac{1}{n})(1-nt-1+n) + f(1)(nt+1-n) - f(t) \\
&= f(1-\tfrac{1}{n})(n-nt) + f(1)(nt+1-n) - f(t)(\underline{1} + \underbrace{n-nt}\ \underbrace{-n+nt}) \\
&= (f(1-\tfrac{1}{n}) - f(t))(n-nt) + (f(1)-f(t))(nt+1-n)
\end{aligned}$$

$\Rightarrow d_p(g_n, f) = (\int_0^1 |f_n(t) - f(t)|^p dt)^{1/p} = (\int_{1-\frac{1}{n}}^1 |(f(1-\frac{1}{n})-f(t))(n-nt) + (f(1)-f(t))(nt+1-n)|^p dt)^{1/p}$

$|a+b|^p \le 2^{p-1}(|a|^p+|b|^p) \quad \le (2^{p-1}\int_{1-\frac{1}{n}}^1 |(f(1-\frac{1}{n})-f(t))(n-nt)|^p + |(f(1)-f(t))(nt+1-n)|^p dt)^{1/p}$

$f \in C[0,1] \Rightarrow \exists M \in R^+$ s.t. $\forall x \in [0,1]$ $|f(x)| \le M$

$|f(x_1) - f(x_2)| \le |f(x_1)| + |f(x_2)| \le 2M$

$\Rightarrow d_p(g_n, f) \le \left\{ 2^{p-1} (2M)^p \left[\int_{1-\frac{1}{n}}^{1} (n-nt)^p \, dt + \int_{1-\frac{1}{n}}^{1} |nt+1-n|^p \, dt \right] \right\}^{\frac{1}{p}}$ $\quad nt+1-n \ge 0$ for $t \ge 1-\frac{1}{n}$

$$= \left\{ 2^{p-1} 2^p M^p \left(\frac{(n-nt)^{p+1}}{p+1} \times -\frac{1}{n} \Big|_{t=1-\frac{1}{n}}^{t=1} + \frac{(nt+1-n)^{p+1}}{p+1} \times \frac{1}{n} \Big|_{t=1-\frac{1}{n}}^{t=1} \right) \right\}^{\frac{1}{p}}$$

$$= \left\{ 2^{2p-1} M^p \left(\frac{1}{p+1}\right) \times \frac{1}{n} (1 - 0 - 0 + 1) \right\}^{\frac{1}{p}}$$

$$= \left\{ \frac{2^{2p} M^p}{p+1} \times \frac{1}{n} \right\}^{\frac{1}{p}} \longrightarrow 0 \text{ as } n \to +\infty, \text{ for any } 1 \le p < \infty.$$

$d_\infty(g_n, f) = \sup \{ |g_n(t) - f(t)| : t \in [0,1] \}$

$= \sup \left| (f(1-\frac{1}{n}) - f(t))(n-nt) + (f(1) - f(t))(nt+1-n) \right| \quad t \in [1-\frac{1}{n}, 1]$

$\le \sup \left| (f(1-\frac{1}{n}) - f(t))(n-nt) \right| + \sup \left| (f(1)-f(t))(nt+1-n) \right| \quad t \in [1-\frac{1}{n}, 1]$

as $n \to +\infty$, $t \to 1$

$\Rightarrow d_\infty(g_n, f) \le \sup | (f(1) - f(1)) \times 0 | + \sup | (f(1) - f(1)) \times 0 |$

$= 0 + 0 = 0$

Problem 34

Let (S, d) be a metric space.

Suppose A, B are two sets in (S, d) and they are open.

By definition, $\forall x \in A$, $\exists \delta > 0$ s.t. $d(x, y) < \delta \Rightarrow y \in A$

$\forall x' \in B$, $\exists \delta' > 0$ s.t. $d(x', y') < \delta' \Rightarrow y' \in B$.

So. $\forall x^* \in A \cup B$

case (I): $x^* \in A \Rightarrow \exists \delta > 0$ s.t. $d(x^*, y^*) < \delta \Rightarrow y^* \in A \Rightarrow y^* \in A \cup B$

case (II): $x^* \in B \Rightarrow \exists \delta' > 0$ s.t. $d(x^*, y^*) < \delta' \Rightarrow y^* \in B \Rightarrow y^* \in A \cup B$

$\Rightarrow \exists \delta^* = \max(\delta, \delta')$ s.t. $d(x^*, y^*) < \delta^* \Rightarrow y^* \in A \cup B$

So $A \cup B$ is open

Similarly. $\forall x^* \in A \cap B \Rightarrow x^* \in A$ and $x^* \in B$

case (I): $x^* \in A \Rightarrow \exists \delta_A \geq 0$ s.t. $d(x^*, y^*) < \delta_A \Rightarrow y^* \in A$

case (II): $x^* \in B \Rightarrow \exists \delta_B > 0$ s.t. $d(x^*, y^*) < \delta_B \Rightarrow y^* \in B$

$\Rightarrow \exists \delta^* = \min(\delta_A, \delta_B) \Rightarrow$ s.t. $d(x^*, y^*) < \delta^* \Rightarrow y^* \in A$ and $y^* \in B$

$\Rightarrow y^* \in A \cap B$

So $A \cap B$ is also open.

Example: $\{ (-\frac{1}{n}, \frac{1}{n}) \}$

Problem 36.

(1) x^n converges to 0 pointwise on $(-1,1)$

proof: $\forall x \in (-1,1)$ for any fixed x. $\lim_{n \to +\infty} x^n = 0 = g(x)$

(2) x^n converges to 0 uniformly on (a,b) where $-1 < a < b < 1$

proof: $\forall x \in (a,b), \forall \varepsilon > 0, |x^n| \leq |x|^n \leq \max\{|a|^n, |b|^n\} (= \varepsilon)$ to find $\underline{n}$

$\exists N_0 \in \mathbb{N}$, which is greater than $\max\left\{ \frac{\log |a|}{\log \varepsilon}, \frac{\log |b|}{\log \varepsilon} \right\}$

$\Rightarrow |x^n - 0| \leq |x|^n \leq \varepsilon$ for all $n \geq N_0$.

So x^n converges to 0 uniformly on (a,b) $-1 < a < b < 1$

(3) x^n does not converge to 0 uniformly on $[0,1)$

By part (1) we know x^n converges to 0 on $(0,1)$ pointwise.

But $|x^n - 0| = x^n > \varepsilon_0 \Rightarrow x > \varepsilon_0^{\frac{1}{n}}$

$\forall n \in \mathbb{N}, \exists \varepsilon_0$. s.t. for any $x > \varepsilon_0^{\frac{1}{n}}$ $|x^n - 0| > \varepsilon_0$

$\Rightarrow x^n$ does not converge to 0 uniformly on $(0,1)$

Problem 1.2

Ω is finite. and $\mathcal{F} = \mathcal{P}(\Omega)$ is a algebra.

So $^{(1)} \Omega \in \mathcal{F}$, $^{(2)} A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$. $^{(3)} A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$.

If $A_n \in \mathcal{F}$ for all $n \geq 1$, and Ω is finite, $|\mathcal{P}(\Omega)| < \infty \Rightarrow$ By induction $\bigcup_{n \geq 1} A_n \in \mathcal{F}$.

So $\mathcal{F}$ is also σ-algebra.

Problem 1.4.

$\forall \theta \in \Theta$, $\mathcal{F}_\theta$ is a σ-algebra.

By definition (1) $\Omega \in \mathcal{F}_\theta$.

(2) If $A \in \mathcal{F}_\theta$, $A^c \in \mathcal{F}_\theta$.

(3). $A, B \in \mathcal{F}_\theta$, $A \cup B \in \mathcal{F}_\theta$

(4) $A_n \in \mathcal{F}_\theta$ for $n \geq 1$ $\bigcup_{n\geq 1} A_n \in \mathcal{F}_\theta$.

$\Rightarrow \forall \theta \in \Theta$, $\Omega \in \mathcal{F}_\theta \Rightarrow \Omega \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta$

$A \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta \Rightarrow A \in \mathcal{F}_\theta$ for any $\theta \in \Theta \Rightarrow A^c \in \mathcal{F}_\theta$ for any $\theta \in \Theta$

$\Rightarrow A^c \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta$

$A, B \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta \Rightarrow A, B \in \mathcal{F}_\theta$ for any $\theta \in \Theta \Rightarrow A \cup B \in \mathcal{F}_\theta$ for any $\theta \in \Theta$

$\Rightarrow A \cup B \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta$

$A_n \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta$ ($n \in \mathbb{N}$) $\Rightarrow A_n \in \mathcal{F}_\theta$ ($n \in \mathbb{N}$) for any $\theta \in \Theta \Rightarrow \bigcup_{n\geq 1} A_n \in \mathcal{F}_\theta$ for any $\theta \in \Theta$

$\Rightarrow \bigcup_{n\geq 1} A_n \in \bigcap_{\theta\in\Theta} \mathcal{F}_\theta$

So $\bigcap_{\theta\in\Theta} \mathcal{F}_\theta$ is also a σ-algebra.

Problem 1.6

$J = \mathbb{N} \Rightarrow \bigcup_{i\in J} A_i = \bigcup_{n\geq 1} A_n = \Omega \Rightarrow \Omega \in \mathcal{F}$.

If $A = \bigcup_{i\in J} A_i \in \mathcal{F}$ for certain $J \subset \mathbb{N}$ $A^c = \Omega \setminus \bigcup_{i\in J} A_i = \bigcup_{i\in J^c} A_i \in \mathcal{F}$, $J^c = \mathbb{N} \setminus J$

If $\{B_n\}_{n\geq 1} \subset \mathcal{F}$, then for every n, $\exists J_n \subset \mathbb{N}$, $B_n = \bigcup_{i\in J_n} A_i$.

$\Rightarrow \bigcup_{n\geq 1} B_n = \bigcup_{i\in J_1} A_i \cup \bigcup_{i\in J_2} A_i \cup \dots = \bigcup_{i\in \bigcup_{n\geq 1} J_n} A_i = \bigcup_{i\in J^*} A_i \in \mathcal{F}$. $\bigcup_{n\geq 1} J_n = J^* \subset \mathbb{N}$.

So. $\mathcal{F}$ is a σ-algebra.

Problem A. Ω = uncountable set

$\mathbb{C}$: collection of all the singleton sets ($\mathbb{C} = \{ \{w\} : w \in \Omega \}$)

class

set with single elements.

$\sigma\langle \mathcal{C} \rangle$ is the smallest σ-algebra generated by $\mathcal{C}$.

proof:

by definition $\sigma\langle \mathcal{C} \rangle = \bigcap_{\mathcal{F} \in I(\mathcal{C})} \mathcal{F}$, $I(\mathcal{C}) \equiv \{ \mathcal{F} : \mathcal{C} \subset \mathcal{F}$ and $\mathcal{F}$ is a σ-algebra on $\Omega \}$

So $\sigma\langle \mathcal{C} \rangle$ is a σ-algebra. since. $\sigma\langle \mathcal{C} \rangle \in I(\mathcal{C})$

And also $\sigma\langle \mathcal{C} \rangle$ is the intersection of all possible σ-algebra containing $\mathcal{C}$.

So $\sigma\langle \mathcal{C} \rangle$ is the smallest σ-algebra.